

Lesson 4-1 · Period 3

Reciprocal Function + Translations — Assessment-Critical

Klimesara-adapted · Student-centered · No Blooket

DOK 1

5 min

bridge from P2

Practice #13 · Generalize k For an inverse variation, write an equation that gives the value of k in terms of x and y .

Then predict: Sketch what you think $y = 1/x$ looks like.
(No wrong answers — just predict. We check in Launch.)

DOK 2

12 min

teacher models

Parent: $f(x) = 1/x$

- ▶ V.A.: $x = 0$ H.A.: $y = 0$
- ▶ Domain: $x \neq 0$ Range: $y \neq 0$
- ▶ Hyperbola, Quadrants I and III. Never touches either axis.

Translated: $g(x) = 1/(x - h) + k$

- ▶ V.A. $\rightarrow x = h$; H.A. $\rightarrow y = k$
- ▶ Domain: $x \neq h$ Range: $y \neq k$
- ▶ (h, k) = **new center** (where asymptotes cross)
- ▶ Ex: $g(x) = 1/(x - 3) + 2 \rightarrow$ asymptotes $x = 3, y = 2$

DOK 2

35 min

student work

6 items total. Parent forms first (laps 1–3), then translations (laps 4–6).

Try It 4 · Scaled parent Graph $g(x) = 10/x$. State domain, range, and asymptotes.

Practice #11 · Error Analysis: $y = 5/x$ Describe and correct the student's graphing error.

Practice #18 · Reflected parent: $y = -2/x$ Graph $y = -2/x$. State domain, range, and asymptotes.

DOK 2

35 min

student work (cont.)

Before each translation: identify (h, k) , state asymptotes, then graph.

Try It 5 · Translated: $g(x) = 1/(x + 2) - 4$ Graph $g(x) = 1/(x + 2) - 4$.
Equations of asymptotes? Domain and range?

Practice #19 · ★ ASSESSMENT CRITICAL $g(x) = 1/(x - 2) + 6$. Asymptotes, domain, range, **and intercepts**. (~8 min)

Practice #21 · Video game storage (if time) Number of games n varies inversely with average game size s (GB).

DOK 2

5 min

DOK-2 reveal + LEHS preview

Sentence frame · Practice #19 pair share “This graph is the parent $f(x) = 1/x$ translated ____ units ____ and ____ units _____. The asymptotes are $x = \underline{\hspace{1cm}}$ and $y = \underline{\hspace{1cm}}$. The intercepts are _____.”

★ **Assessment Signal — Practice #19 is the KEYSTONE RE-HEARSAL** Practice #19 is the keystone rehearsal item for Topic 4 LEHS:

- ▶ **Q#3** (H/V asymptotes) — #19 asked for exactly this.
- ▶ **Q#5** (translation description) — the sentence frame above IS the answer shape.
- ▶ #19 hit BOTH skills in one item. If you got #19, you are ready.

Exit Ticket — Pre-Assessment Confidence Check

Algebra 2 · Lesson 4-1 · Period 3

DOK 1-2

3 min

not graded

Complete on your exit ticket

1. I can identify asymptotes of $y = 1/(x - h) + k$. Confidence: 1 / 2 / 3 / 4 / 5
2. I can describe a translation from $f(x) = 1/x$ to $g(x)$. Confidence: 1 / 2 / 3 / 4 / 5
3. One thing I want to review before Topic 4 assessment:
_____.